

30

Multimodal AI Interview Questions

**Vision, documents, voice, and video —
the round most candidates skip preparing.**

How models actually see · ColPali retrieval · voice agents ·
visual injection · one question per page.

Naved Khan
Senior Gen-AI Engineer

READ THIS FIRST

The model doesn't see pictures. It reads them, tile by tile.

Once you know images become tokens, everything follows: why vision costs money, why tables misread, why injection hides in pixels — and why every discipline you learned in text transfers straight across.

“How does a language model actually “see” an image?”

— THE FULL ANSWER

- ▶ The image gets cut into a grid of small squares — **patches** — and each patch becomes a token, like a word.
- ▶ Those patch tokens are projected into the **same space as text tokens**, so the model reads an image the way it reads a sentence.
- ▶ That's why vision costs money: a detailed image can be **hundreds to thousands of tokens**.
- ▶ Plain words: the model doesn't see pictures. It reads them, tile by tile.

“What does an image cost per request?”

— THE FULL ANSWER

- ▶ It depends on resolution: images are billed as **tokens**, and more detail means more patch tiles.
- ▶ Most APIs offer a **detail setting**: low detail costs a fixed small amount; high detail scales with size.
- ▶ The lever everyone forgets: **resize before sending**. A 4K screenshot of a button costs 10× what the button needs.
- ▶ Say the habit: "images go through a resize-and-detail policy, not raw from the user's phone."

```
messages=[{"type": "image_url", "detail": "low",  
...}] # cost lever
```

“Native multimodal model vs a vision model bolted onto an LLM — difference?”

— THE FULL ANSWER

- ▶ **Native:** one model trained on text and images together, reasoning across both in one pass.
- ▶ **Bolted-on:** a separate vision encoder describes the image, and the LLM works from that description — details die in translation.
- ▶ In 2026 the frontier models are native, and it shows most on **charts, tables, and spatial questions**.
- ▶ The interview line: "captioning pipelines lose exactly the details the question usually needs."

“When does a vision model beat OCR, and when does OCR still win?”

— THE FULL ANSWER

- ▶ Vision wins when **layout carries meaning**: tables, charts, forms, diagrams, handwriting. OCR flattens those into word soup.
- ▶ OCR still wins for **bulk plain text**: it's cheaper, faster, deterministic, and searchable at scale.
- ▶ The production pattern is both: OCR for the text layer, vision for the pages where **structure is the content**.
- ▶ Never say "OCR is dead" — say "OCR is a cost tier."

“Can the model hallucinate what's in an image?”

— THE FULL ANSWER

- ▶ Yes, constantly — it reads plausibility, not pixels. Ask about a serial number and it may **produce one that looks right**.
- ▶ Highest-risk zones: small text, numbers in dense tables, things **partially cropped** out of frame.
- ▶ Mitigations: higher detail settings for text-heavy images, asking the model to **quote exactly what it sees**, and verification for anything that matters.
- ▶ Same rule as text: grounded, cited, allowed to say "can't read it."

“What's genuinely hard in multimodal that's easy in text?”

— THE FULL ANSWER

- ▶ **Evals.** Text answers can be string-matched; "did it read the chart correctly" needs labeled images and judgment.
- ▶ **Chunking has no obvious equivalent** — you can't split an image mid-table and expect meaning to survive.
- ▶ **Cost surprises:** one image can outweigh a thousand words of text, literally, in tokens.
- ▶ Naming these three shows you've built with it, not just demoed it.

“Design document understanding for 10,000 scanned PDFs.”

— THE FULL ANSWER

- ▶ Triage first: **born-digital pages get text extraction** (cheap, perfect); scanned pages get OCR; layout-heavy pages get vision.
- ▶ Store both layers: extracted text for search, **page images for the hard questions**.
- ▶ Tables and charts route to the vision model with high detail — that's where extraction quietly lies.
- ▶ The senior sentence: "one pipeline, three cost tiers, routed per page — not one tool for everything."

“What is ColPali-style retrieval, and why did it matter?”

— THE FULL ANSWER

- ▶ It **skips OCR entirely**: each document page is embedded directly as an image, patch by patch, and queries match against those visual patches.
- ▶ That preserves what OCR destroys: **tables, charts, diagrams, layout** — the page IS the representation.
- ▶ Cost: heavier index (many vectors per page) and late-interaction scoring — you pay in storage and compute.
- ▶ One line: "retrieval that reads pages the way humans skim them — visually."

“The model keeps misreading numbers in dense tables. Debug it.”

— THE FULL ANSWER

- ▶ First check the **input resolution** — most table failures are just downscaled images where digits blurred together.
- ▶ Then crop: send **the table region alone** at high detail instead of the whole page.
- ▶ Then verify: extract twice (vision + OCR) and **flag disagreements** for review — cheap and catches most damage.
- ▶ The habit being tested: you debug the pixels before blaming the model.

“A user uploads a chart and asks “what's the trend?” What can go wrong?”

— THE FULL ANSWER

- ▶ Truncated axes, log scales, and dual axes fool models the **same way they fool humans** — but with total confidence.
- ▶ Legends far from lines cause **series mix-ups**: the model says revenue when it read costs.
- ▶ Mitigation: ask it to first **describe the axes and legend**, then answer — a chain-of-sight before the conclusion.
- ▶ That two-step pattern fixes a surprising share of chart errors, and it's free.

“Forms and invoices: extraction pipeline in three sentences.”

— THE FULL ANSWER

- ▶ Vision model reads the page into a **strict schema** — schema-enforced output, never free text.
- ▶ Validators check the arithmetic: totals sum, dates parse, **tax math balances** — deterministic checks on model output.
- ▶ Below a confidence bar, the field routes to a **human queue**; every correction becomes training and eval data.
- ▶ Extraction is where multimodal meets your structured-output discipline.

```
Invoice.model_validate_json(reply)
assert inv.total == sum(l.amount for l in inv.lines)
```

“Screenshots as a data source: what changes?”

— THE FULL ANSWER

- ▶ Screenshots are the **fastest-growing input type**: bug reports, dashboards, app states — context nobody types out.
- ▶ They're layout-heavy by nature, so **vision-first**, and they're where visual agents (reading UIs) begin.
- ▶ Watch the privacy edge: screenshots carry **whatever was on screen** — names, emails, tokens. Scrub like any input.
- ▶ "A screenshot is a document with zero structure and maximum context" is a good line to own.

“Design multimodal RAG: text and images in one corpus. Two architectures.”

— THE FULL ANSWER

- ▶ **Caption-and-embed**: describe every image in text, index the text. Simple, cheap, works with your existing stack — loses visual detail.
- ▶ **Native multimodal embeddings**: images and text in **one vector space** (CLIP-style), retrieved together. Truer, heavier.
- ▶ Start with captions; upgrade the slices where detail loss hurts — usually **charts and product images**.
- ▶ The trade-off named out loud is the answer; picking one "because it's better" isn't.

“Explain CLIP-style embeddings in plain words.”

— THE FULL ANSWER

- ▶ One model learns to place **images and their descriptions close together** in the same vector space.
- ▶ So "red running shoes" the text lands near photos of red running shoes — **search across modalities** with one query.
- ▶ Trained on hundreds of millions of image-caption pairs; the alignment IS the product.
- ▶ It's the reason "find me images like this sentence" stopped being science fiction.

“How do you evaluate multimodal retrieval?”

— THE FULL ANSWER

- ▶ Same discipline, new labels: a golden set of **real queries mapped to the right pages/images**, recall@k tracked per change.
- ▶ Add a **modality split**: text queries → image answers, and the reverse, measured separately — they fail differently.
- ▶ Judge the final answers with a **vision-capable judge**, calibrated on human labels.
- ▶ Benchmarks exist, but your corpus is the exam that counts — same rule as text RAG.

“When is multimodal RAG the wrong answer?”

— THE FULL ANSWER

- ▶ When 95% of questions hit **plain text** — captions or extraction cover the rest at a fraction of the cost.
- ▶ When the images are **decorative**, not informational: logos, stock photos, headshots.
- ▶ When latency is tight: multimodal retrieval and vision calls are **slower per hop**.
- ▶ "Not yet" earns the same respect here as everywhere else in this series.

“Mixed corpus: how do you keep image and text results comparable?”

— THE FULL ANSWER

- ▶ Don't trust raw similarity scores **across different embedding models** — they're not on the same scale.
- ▶ Retrieve per modality, then **rerank together** with a model that sees both, or fuse ranks (RRF) instead of scores.
- ▶ Metadata carries the modality tag, so the answer can **cite "page 4, chart"** and not just "chunk 7".
- ▶ Rank fusion over score fusion — the same trick as hybrid search, one level up.

“One image in the context — where does it go and what does it displace?”

— THE FULL ANSWER

- ▶ Images are **token-expensive residents** of the same context budget: one high-detail page can cost 1,000+ tokens.
- ▶ So the layout rule extends: images the question needs go in; "maybe useful" images stay **retrievable but out**.
- ▶ Refer to them by ID, fetch on demand — an image scratchpad, same as text compaction.
- ▶ "Every token must earn its place" survives the modality change untouched.

“Voice agents: pipeline vs speech-to-speech. Trade-offs.”

— THE FULL ANSWER

- ▶ **Pipeline** (speech→text→LLM→speech): controllable, debuggable, each stage swappable — but latency stacks per hop.
- ▶ **Native speech-to-speech**: sub-200ms responses that feel human, prosody preserved — but a harder box to control and eval.
- ▶ 2026 default: speech-to-speech for conversation, **pipeline for anything regulated** where you must log exact text.
- ▶ Latency is the product in voice: 500ms of silence feels broken in a way chat never does.

“What breaks in voice that never breaks in chat?”

— THE FULL ANSWER

- ▶ **Interruptions:** humans barge in mid-sentence; the agent must stop, listen, and recover state.
- ▶ **No retry button:** users won't repeat themselves twice — transcription errors compound into wrong actions.
- ▶ Accents, noise, and names: the **error rate is demographic**, and that's a fairness bug, not a nit.
- ▶ Design answer: confirmations before actions, and "did I get that right?" as a first-class pattern.

“Video understanding: how does it actually work today?”

— THE FULL ANSWER

- ▶ Mostly **frame sampling**: pull frames every N seconds, treat them as an image sequence with timestamps, plus the audio transcript.
- ▶ Dense sampling for action-heavy video, sparse for talking heads — **sampling policy is the cost dial**.
- ▶ True native video models are arriving, but sampled-frames-plus-transcript remains the production workhorse.
- ▶ The honest line: "most video AI is image AI with a clock."

```
frames = [clip.get_frame(t) for t in range(0, dur,  
5)] # 1 frame / 5s
```

“Build meeting notes from a recording. Sketch it.”

— THE FULL ANSWER

- ▶ Transcribe with **speaker separation**, then chunk by topic shifts, not by minutes.
- ▶ Summarize per section with the transcript as evidence — **quotes linked to timestamps**, so claims are verifiable.
- ▶ Action items go through **structured extraction**: owner, task, deadline — validated, not vibes.
- ▶ The killer feature is trust: every line of the summary **clicks back to the moment it came from**.

“How do you eval a voice agent end to end?”

— THE FULL ANSWER

- ▶ Three layers: **transcription accuracy** (word error rate on your accents, not lab audio), **task success** on scripted scenarios, and **latency** at p95.
- ▶ Simulated callers with personas and background noise make it repeatable.
- ▶ And one metric chat never needed: **interruption recovery rate** — does barging in break it?
- ▶ Voice evals are theater rehearsals, not unit tests. Budget accordingly.

“Multimodal latency: where does the time go?”

— THE FULL ANSWER

- ▶ Image encoding adds **hundreds of ms per high-detail image** before generation even starts.
- ▶ Voice adds capture, transcription (or native processing), and synthesis on the way out.
- ▶ Levers: resize aggressively, lower detail where text isn't tiny, **parallelize vision calls**, stream everything.
- ▶ Same law as text serving: set the budget first, spend inside it.

“What does visual prompt injection look like?”

— THE FULL ANSWER

- ▶ Instructions **hidden inside images**: text in a screenshot, a sign in a photo, white-on-white words the model reads and obeys.
- ▶ It's the indirect injection story again — but now the payload rides in pixels, where **text filters can't see it**.
- ▶ Defense is the same blast-radius stack: images are data, tools are allowlisted, actions gated.
- ▶ Say it plainly: "anything the model can read is an input channel — including what's written inside a picture."

“PII in images: what changes about privacy?”

— THE FULL ANSWER

- ▶ Images leak differently: **faces, screens in the background, documents on desks**, badge photos — things no text scrubber sees.
- ▶ Detection needs vision: face and text-region detectors at ingestion, **before** anything is stored or indexed.
- ▶ Retention gets stricter: a stored screenshot is a stored everything-that-was-on-screen.
- ▶ Rule of thumb: treat every uploaded image as a document you haven't read yet — because it is.

“Pick a vision model for production. What's the checklist?”

— THE FULL ANSWER

- ▶ Same as text, one addition: **your golden set now contains images** — your charts, your forms, your lighting.
- ▶ Check the pricing model for images specifically: **per-tile costs differ wildly** between providers.
- ▶ Latency per image at your detail settings, not the demo's.
- ▶ The closer: "the best vision model is the one that passes MY image suite at the lowest cost — benchmarks don't include our invoices."

“Multimodal in agents: what does vision unlock?”

— THE FULL ANSWER

- ▶ **Computer use:** agents that read screens and click — vision is the eyes of every UI-operating agent.
- ▶ Richer inputs: a user photographs the broken part instead of describing it; the agent sees the error dialog directly.
- ▶ New failure modes too: **UI changes break visual agents** like API changes break integrations.
- ▶ One sentence: vision turned agents from text processors into **operators**.

“Where do multimodal costs surprise teams?”

— THE FULL ANSWER

- ▶ **Image tokens dwarf text tokens:** one screenshot-heavy conversation can cost more than fifty text chats.
- ▶ Video is quadratically dangerous: frames × detail × length — **sampling policy is the bill.**
- ▶ Storage grows too: page images, frame caches, audio files — infra costs text never had.
- ▶ The defense is familiar: resize policies, detail tiers, caching, and **cost per successful task** on the dashboard.

“Close it out: is multimodal a separate skill, or the same job?”

— THE FULL ANSWER

- ▶ Same job, new inputs. Every discipline transfers: **golden sets, budgets, blast radius, "not yet"** — the nouns changed, the verbs didn't.
- ▶ What's genuinely new: eval labeling is harder, costs hide in pixels, and injection rides in images.
- ▶ The senior close: "I treat modalities as cost tiers and input channels — **the engineering discipline underneath doesn't change.**"
- ▶ That sentence ends the round. Everything in this deck is its proof.

Preparing for AI interviews?



Naved Khan

Senior Gen-AI Engineer

This completes the set: **RAG, agents, evals, inference, MCP, safety, context engineering** — **and now multimodal**. All on my profile.